resulting edge response. The 10%-90% distance of this curve is the best single parameter measurement of the system's resolution. To keep your boss and the marketing people happy, take the first difference of the edge response to find the LSF, and then use the FFT to find the MTF.

Sample Spacing and Sampling Aperture

Figure 25-6 shows two extreme examples of sampling, which we will call a **perfect detector** and a **blurry detector**. Imagine (a) being the surface of an imaging detector, such as a CCD. Light striking anywhere inside one of the square pixels will contribute *only* to that pixel value, and no others. This is shown in the figure by the black sampling aperture exactly filling one of the square pixels. This is an optimal situation for an image detector, because *all* of the light is detected, and there is *no* overlap or crosstalk between adjacent pixels. In other words, the sampling aperture is exactly equal to the sample spacing.

The alternative example is portrayed in (e). The sampling aperture is considerably larger than the sample spacing, and it follows a Gaussian distribution. In other words, each pixel in the detector receives a contribution from light striking the detector in a region *around* the pixel. This should sound familiar, because it is the output side viewpoint of convolution. From the corresponding input side viewpoint, a narrow beam of light striking the detector would contribute to the value of several neighboring pixels, also according to the Gaussian distribution.

Now turn your attention to the edge responses of the two examples. The markers in each graph indicate the actual pixel values you would find in an image, while the connecting lines show the *underlying curve* that is being sampled. An important concept is that the shape of this underlying curve is determined *only* by the *sampling aperture*. This means that the resolution in the final image can be limited in two ways. First, the underlying curve may have poor resolution, resulting from the sampling aperture being too large. Second, the sample spacing may be too large, resulting in small details being lost between the samples. Two edge response curves are presented for each example, illustrating that the actual samples can fall anywhere along the underlying curve. In other words, the edge being imaged may be sitting exactly upon a pixel, or be straddling two pixels. Notice that the perfect detector has *zero* or *one* sample on the rising part of the edge. Likewise, the blurry detector has *three* to *four* samples on the rising part of the edge.

What is limiting the resolution in these two systems? The answer is provided by the *sampling theorem*. As discussed in Chapter 3, sampling captures all frequency components below one-half of the sampling rate, while higher frequencies are lost due to aliasing. Now look at the MTF curve in (h). The sampling aperture of the blurry detector has removed all frequencies greater than one-half the sampling rate; therefore, *nothing* is lost during sampling. This means that the resolution of this system is